# Mika Vohl

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## Education

**McGill University** September 2023 – May 2028
*Bachelor of Science, Joint Honours in Computer Science and Physics* *Montreal, Canada*

- **Cumulative GPA:** 3.93/4.0 - Dean's Honour List 2023/2024
- **President - Hack4Impact:** Led a 45+ member student collective to build full-stack applications for nonprofits
- **Developer - McHacks:** Built and maintained software to operate McGill's largest hackathon (500+ users)

## Technical Skills

**Languages**: Python, Go, C++, Java, C, JavaScript, TypeScript, SQL, Bash, HTML/CSS, PHP
**Frameworks & Libraries**: PyTorch, CUDA, ROCm, Flask, React, Tensorflow/Keras, Matplotlib, Pandas, scikit-learn
**Developer Tools**: Git, Docker, Kubernetes, Linux, Github Actions, AWS, Hugging Face, PostgreSQL, llama.cpp
**Certifications**: Stanford Machine Learning Specialization, AMD HIP GPU Programming

## Experience

**Machine Learning Libraries Intern** May 2026 – Present
*AMD* *Calgary, Canada*

- Improved kernel selection by 4.3% by building an ML-based pipeline to tune AMD's GEMM latency model
- Automated GPU profiling during GEMM tuning to identify kernel-level performance bottlenecks
- Tuned GEMM kernels for OpenAI, Meta, and Mistral AI models on AMD GPUs, reducing latency by up to 10%
- Shipped features and improvements to AMD's open-source GPU libraries serving thousands of enterprise users

**Researcher** September 2025 – July 2026
*McGill AI Society* *Montreal, Canada*

- Researching deep learning on missing clinical data; supervised by Mila grad student; accepted to AI4PH poster fair
- Benchmarked TabPFN and ConTextTab models against MLP/XGBoost on predicting mortality under missingness

**Software Engineer Intern (Machine Learning)** May 2025 – Aug. 2025
*streamwise.gg* *Montreal, Canada*

- Fine-tuned an open-source Transducer model for ASR in livestreams, cutting transcription error from 26% to 10%
- Built a low-latency audio/video pipeline using Go and FFmpeg, handling transcoding, buffering, multiqueueing
- Containerized and orchestrated services with Docker & Kubernetes, managing networking, scaling, and lifecycles

**Software Engineer Intern** May 2024 – Sept. 2024
*myBonum* *Toronto, Canada*

- Built an API for interfacing with an SQL database, enabling core backend functionality for a healthcare startup
- Integrated Stripe billing and subscriptions directly into user-facing dashboard and backend infrastructure

**Software Developer Intern** June 2023 – Sept. 2023
*CAIL* *Toronto, Canada*

- Shipped features and platform support for an HP NonStop terminal emulator, including Linux and Android
- Improved terminal throughput by 80% through rendering, sorting, and I/O optimizations

## Projects

**Predicting Chaos with Convolutional Neural Nets** - Try it out! | *Python, PyTorch, Typescript* May 2025

- Trained a CNN to predict the 5th next state of Conway's Game of Life, a chaotic system, with 98% accuracy
- Built a custom Game of Life engine to generate 50,000 synthetic board states for training a convolutional network

**Stitch** - 1st place winner at McGill AI Society Hacks 25-26 - GitHub | *Typescript, React, Python* November 2025

- Built a drag-and-drop platform to design, train, and test neural networks for handwritten digit recognition
- Provided users end-to-end control of real PyTorch models by dynamically compiling and executing custom model instances from user specifications, streaming live results back to the user via Server-Sent Events

**PyTorch-Style Tensor Autograd Engine** - GitHub | *Python, Numpy* June 2025

- Built an autograd engine for propagating gradients across ML computations, supporting tensor-wise operations
- Significantly accelerated neural network computations using Numpy's BLAS vectorized operations